

Problem 2

Maksym Stetskov, Mykyta Huskov

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PROBLEM 2

PART 2: Exploratory Data Analysis

Imports

```
library(here)
library(tidyverse)
library(stargazer)
library(ggplot2)
library(zoo)
library(readxl)
library(stringr)

library(lmtest)
library(sandwich)
library(tseries)
library(knitr)
library(tinytex)
```

Load the main dataset

Capture original column names. Drop the time variable.

Rename columns for convinience

2.2 Adding new variables from CNB and Eurostat data

Import CNB and Eurostat sources.

```
cnb_loans_interest <- read_excel(here("hw1", "data", "cnb_loans_interest.xlsx"), col_names = FALSE, skip = 1)
cnb_loans_volume <- read_excel(here("hw1", "data", "cnb_loans_volume.xlsx"), col_names = FALSE, skip = 1)
eurostat_prod_prices <- read_excel(here("hw1", "data", "eurostat_prod_prices.xlsx"), col_names = FALSE, skip = 1)
eurostat_employed <- read_excel(here("hw1", "data", "eurostat_employed.xlsx"), col_names = FALSE, sheet = "Employment")
```

Normalize quarter formatting

```
data_houses$year_quarter <- as.yearqtr(gsub(" - ", " ", data_houses$year_quarter))
```

Extract relevant columns from CNB data

```

cnb_loans_interest <- cnb_loans_interest[2:41, c(1, 2)] %>%
  rename_with(~ c("yr_qr", "house_loan_interest")) %>%
  arrange(yr_qr)
cnb_loans_volume <- cnb_loans_volume[2:41, c(1, 2)] %>%
  rename_with(~ c("yr_qr", "house_loan_volume")) %>%
  arrange(yr_qr)

```

Attach CNB series to houses

```

data_houses$house_loan_interest <- parse_double(cnb_loans_interest$house_loan_interest)
data_houses$house_loan_volume <- parse_double(cnb_loans_volume$house_loan_volume)

```

Define Eurostat import helper

```

eurostat_import <- function(eurostat_df, col_name = "value") {
  # Pull time and value rows
  time <- as.vector(t(eurostat_df[1, 2:ncol(eurostat_df)]))
  vals <- as.vector(t(eurostat_df[3, 2:ncol(eurostat_df)]))

  # Build output df
  out <- data.frame(yr_qr = time, value = vals) %>%
    dplyr::mutate(
      yr_qr = gsub("-", " ", yr_qr),
      yr_qr = as.yearqtr(yr_qr, format = "%Y Q%q"),
      value = na_if(value, ":"),
      value = parse_double(value)
    ) %>%
    tidyr::drop_na() %>%
    dplyr::filter(
      yr_qr >= as.yearqtr("2015 Q1", format = "%Y Q%q"),
      yr_qr <= as.yearqtr("2024 Q4", format = "%Y Q%q")
    ) %>%
    arrange(yr_qr)

  # Rename value to target name
  colnames(out)[2] <- col_name
  return(out)
}

```

Import Eurostat employed and producer prices series

```

eurostat_employed <- eurostat_import(eurostat_employed, col_name = "employed_1000")
eurostat_prod_prices <- eurostat_import(eurostat_prod_prices, col_name = "prod_price_idx_2021")

```

Append Eurostat series to houses.

```

data_houses$employed_1000 <- eurostat_employed$employed_1000
data_houses$prod_price_idx_2021 <- eurostat_prod_prices$prod_price_idx_2021

```

Adding invasion of Ukraine and quarter dummies

```

data_houses <- data_houses %>%
  mutate(
    ones = 1L,
    quart = str_sub(as.character(year_quarter), -2),
  ) %>%
  pivot_wider(
    names_from = quart,
    values_from = ones,
    values_fill = 0,
    values_fn = sum
  )

data_houses <- data_houses %>%
  mutate(ukr_invasion = as.integer(year_quarter >= as.yearqtr("2022 Q1")))

```

Cleaning intermediary objects

```

keep <- c("data_houses", "true_names")
rm(list = setdiff(ls(envir = .GlobalEnv), keep), envir = .GlobalEnv)

```

2.2.3 Summary table of the data —

```

summary_stats <- function(df_col) {
  stats <- c(
    min(df_col, na.rm = TRUE),
    quantile(df_col, 0.25, na.rm = TRUE),
    median(df_col, na.rm = TRUE),
    mean(df_col, na.rm = TRUE),
    quantile(df_col, 0.75, na.rm = TRUE),
    max(df_col, na.rm = TRUE),
    sum(is.na(df_col))
  )
  return(stats)
}

```

```
print(data_houses[10], n=40)
```

```

## # A tibble: 40 x 1
##   cpi_yr_per
##   <dbl>
## 1      0.1
## 2     0.67
## 3     0.37
## 4      0.1
## 5     0.47
## 6     0.27
## 7     0.53
## 8     1.47
## 9     2.43
## 10    2.22

```

```
## 11      2.55
## 12      2.6
## 13      1.85
## 14      2.27
## 15      2.39
## 16      2.09
## 17      2.71
## 18      2.82
## 19      2.84
## 20      3.02
## 21      3.6
## 22      3.11
## 23      3.31
## 24      2.63
## 25      2.19
## 26      2.9
## 27      4.12
## 28      6.13
## 29     11.2
## 30     15.8
## 31     17.6
## 32     15.7
## 33     16.4
## 34     11.1
## 35      8.04
## 36      7.56
## 37      2.09
## 38      2.46
## 39      2.31
## 40      2.88
```

Initialize summary table header.

```
summary_df <- data.frame(
  "Summary Statistics" = c(
    "Min", "First Quartile", "Median", "Mean", "Third Quartile", "Max",
    "Number of NAs"
  )
)
```

Extend display names list.

```
true_names <- c(true_names, "Loans on house purchase (millions of CZK, current prices)", "Interest on h
```

Pair display names with sources.

```
paired_names <- Map(list, true_names, colnames(data_houses)[-1])
```

```
## Warning in mapply(FUN = f, ..., SIMPLIFY = FALSE): longer argument not a
## multiple of length of shorter
```

Fill summary table columns.

```

for (cols in paired_names) {
  new_name <- cols[[1]]
  old_name <- cols[[2]]

  df_col <- data_houses[[old_name]]

  if (is.numeric(df_col)) {
    summary_df[[old_name]] <- summary_stats(df_col)
  }
}

```

2.2.4 Visualizations of the most important variables —

Time series plot with non-linear trend

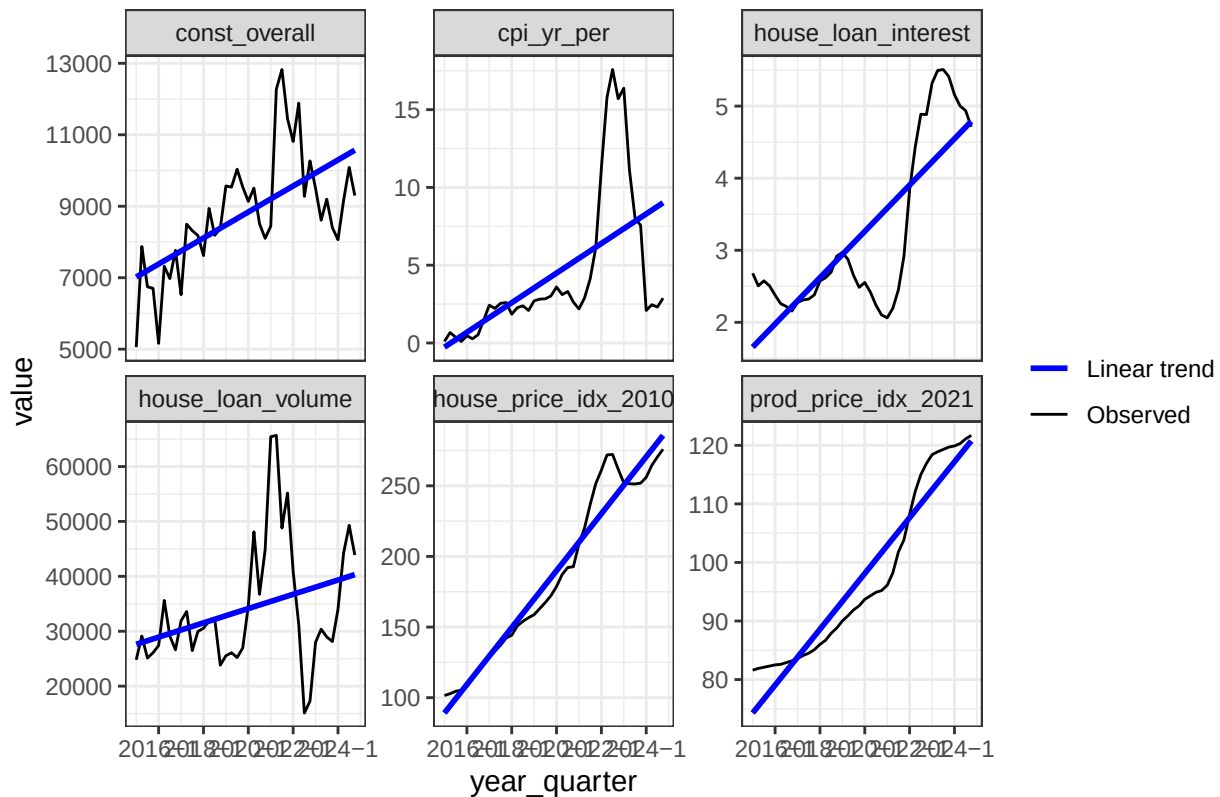
```

data_houses %>%
  pivot_longer(
    cols = c(house_price_idx_2010, cpi_yr_per, house_loan_interest, const_overall, house_loan_volume, p
    names_to = "variable",
    values_to = "value"
  ) %>%
  ggplot(aes(x = year_quarter, y = value, group = variable)) +
  geom_line(aes(color = "Observed")) +
  geom_smooth(aes(color = "Linear trend"), method = "lm", se = FALSE) +
  facet_wrap(~variable, scales = "free_y") +
  scale_color_manual(values = c("Observed" = "black", "Linear trend" = "blue")) +
  guides(color = guide_legend(title = NULL)) +
  theme_bw() +
  labs(
    title = "Time series of the most important variables"
  )

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Time series of the most important variables



2.2.5 Stationarity and trending variables —

```
data_houses %>%
  pivot_longer(
    cols = -c(year_quarter, Q1, Q2, Q3, Q4, ukr_invasion),
    names_to = "variable", values_to = "value"
  ) %>%
  ggplot(aes(x = year_quarter, y = value, group = variable)) +
  geom_vline(
    aes(
      xintercept = as.yearqtr("2022 Q1"),
      color = "Ukraine invasion"
    ),
    linewidth = 1.1
  ) +
  geom_line(aes(color = "Observed"), alpha = 0.9) +
  geom_smooth(aes(color = "Non-linear trend"), method = "loess", se = FALSE) +
  geom_point(aes(color = "Observed"), size = 0.8) +
  facet_wrap(~variable, scales = "free_y") +
  theme_classic() +
  scale_color_manual(
    values = c(
      "Observed" = "black",
      "Non-linear trend" = "blue",

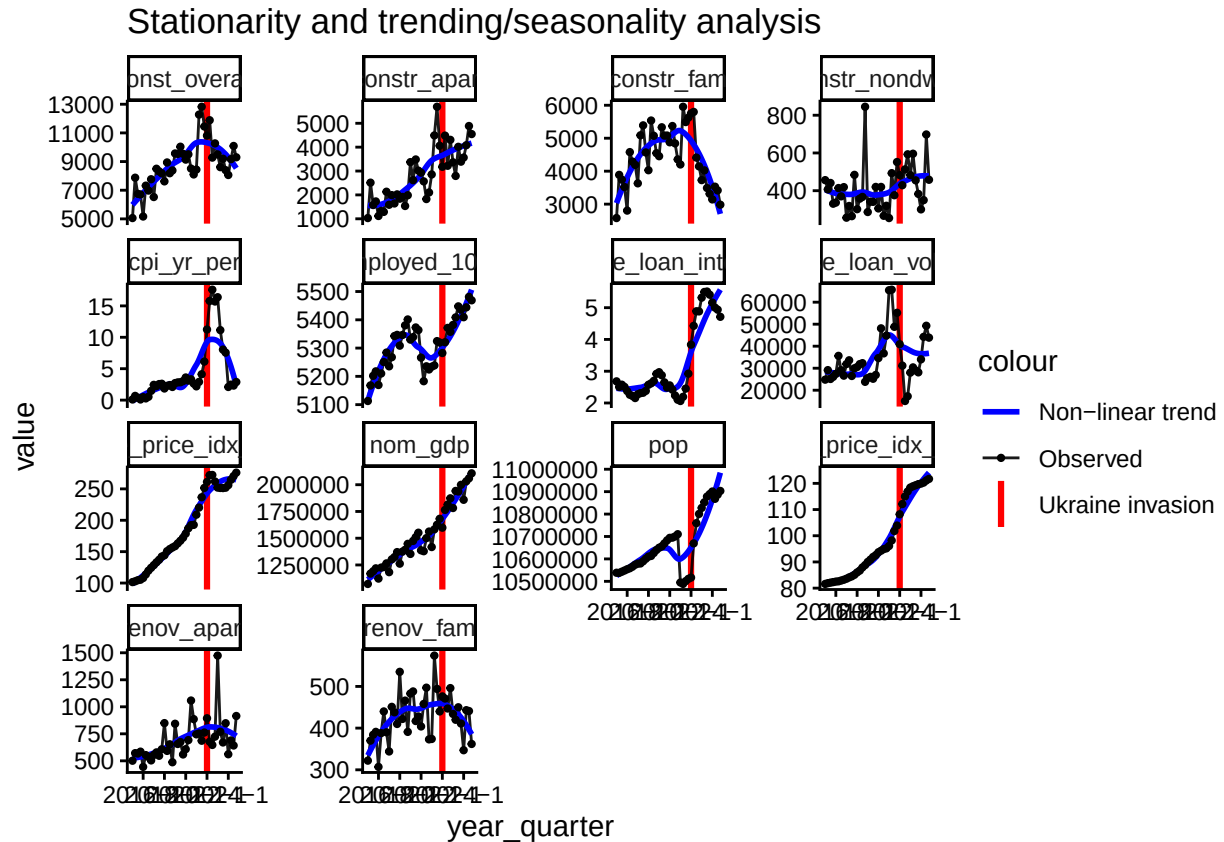
```

```

    "Ukraine invasion" = "red"
  )
) +
labs(
  title = "Stationarity and trending/seasonality analysis"
)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Part 3 —

2.3.1 —

```

sm<-lm(data_houses$house_price_idx_2010~data_houses$nom_gdp)
summary(sm)

```

```

##
## Call:
## lm(formula = data_houses$house_price_idx_2010 ~ data_houses$nom_gdp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max

```

```
## -27.647 -16.099 -7.743 10.541 59.162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.080e+02  1.847e+01  -5.846 9.27e-07 ***
## data_houses$nom_gdp  1.939e-04  1.191e-05  16.274 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.67 on 38 degrees of freedom
## Multiple R-squared:  0.8745, Adjusted R-squared:  0.8712
## F-statistic: 264.8 on 1 and 38 DF,  p-value: < 2.2e-16
```

```
res_sm<-residuals(sm)
bp_sm<-lm(res_sm^2~data_houses$nom_gdp)
bptest(bp_sm)
```

```
##
## studentized Breusch-Pagan test
##
## data:  bp_sm
## BP = 0.14552, df = 1, p-value = 0.7028
```

```
bgtest(sm)
```

```
##
## Breusch-Godfrey test for serial correlation of order up to 1
##
## data:  sm
## LM test = 17.49, df = 1, p-value = 2.888e-05
```

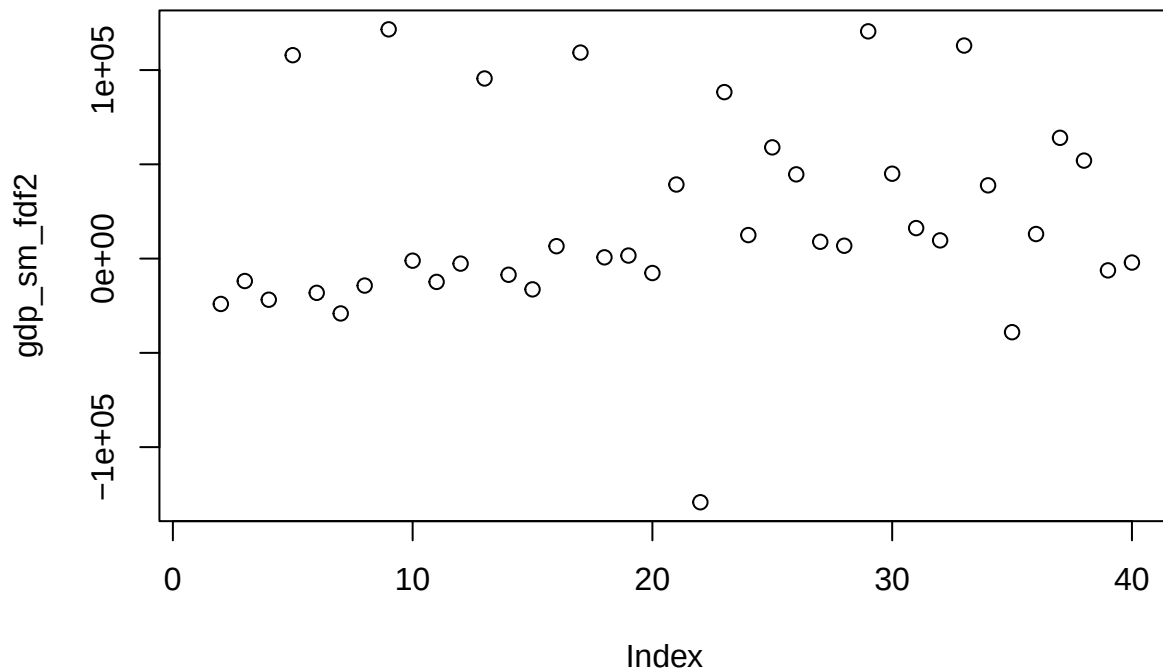
```
adf.test(data_houses$nom_gdp)
```

```
##
## Augmented Dickey-Fuller Test
##
## data:  data_houses$nom_gdp
## Dickey-Fuller = -0.75762, Lag order = 3, p-value = 0.9569
## alternative hypothesis: stationary
```

```
quarter <- substr(data_houses$year_quarter, 7, 7)
print(quarter)
```

```
## [1] "1" "2" "3" "4" "1" "2" "3" "4" "1" "2" "3" "4" "1" "2" "3" "4" "1" "2" "3"
## [20] "4" "1" "2" "3" "4" "1" "2" "3" "4" "1" "2" "3" "4" "1" "2" "3" "4" "1" "2"
## [39] "3" "4"
```

```
gdp_sm_fdf <- lm(formula = data_houses$nom_gdp ~ as.factor(quarter), data = data_houses)
gdp_sm_fdf1 <- gdp_sm_fdf$residuals
gdp_sm_fdf2 <- c(NA,diff(gdp_sm_fdf1))
plot(gdp_sm_fdf2)
```



```
adf.test(data_houses$house_price_idx_2010)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: data_houses$house_price_idx_2010
## Dickey-Fuller = -2.7787, Lag order = 3, p-value = 0.2679
## alternative hypothesis: stationary
```

2.3.2 —

```
gdp_detrend_ts <- ts(data_houses$nom_gdp, start = c(2015, 1), frequency = 4)
gdp_detrend1 <- diff(gdp_detrend_ts, lag = 4)
print(gdp_detrend1)
```

```
##      Qtr1  Qtr2  Qtr3  Qtr4
## 2016 50102 56009 38805 46301
## 2017 60004 77065 93798 105447
## 2018 79385 71905 67947 77192
## 2019 90935 100175 118107 103879
## 2020 33903 -95990 -9295 10831
## 2021 30534 204440 125061 119402
## 2022 180932 181278 188536 191371
```

```
## 2023 183844 177671 122448 125820
## 2024 76882 90009 122804 107682
```

```
hpi_detrend_ts <- ts(data_houses$house_price_idx_2010, start = c(2015, 1), frequency = 4)
hpi_detrend <- diff(hpi_detrend_ts, lag = 4)
print(hpi_detrend)
```

```
##      Qtr1 Qtr2 Qtr3 Qtr4
## 2016   6.5 10.9 15.2 19.0
## 2017  20.5 19.3 17.4 17.9
## 2018  15.7 17.8 17.0 14.4
## 2019  14.6 12.3 13.3 15.6
## 2020  19.9 24.1 24.8 20.5
## 2021  30.2 33.0 44.5 58.9
## 2022  52.2 51.7 35.5 10.2
## 2023  -9.0 -20.4 -20.9 -9.9
## 2024   3.9 13.0 19.2 24.0
```

2.3.3 —

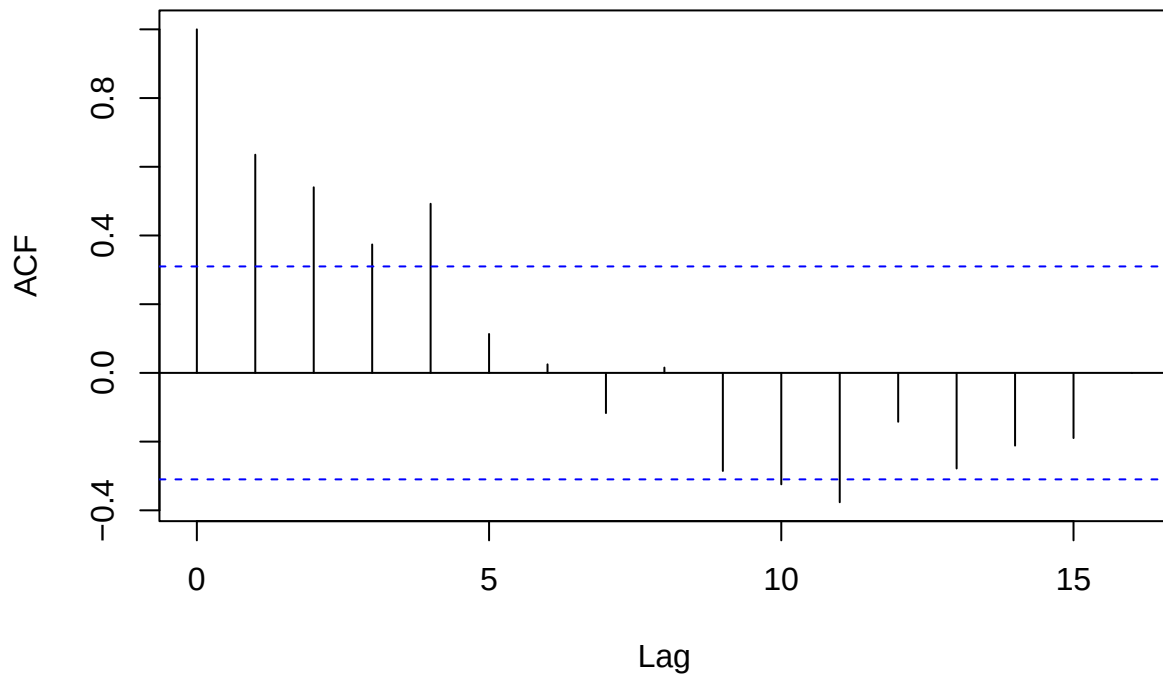
```
sm_fdf <- lm(hpi_detrend_ts ~ gdp_detrend_ts)
summary(sm_fdf)
```

```
##
## Call:
## lm(formula = hpi_detrend_ts ~ gdp_detrend_ts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -27.647 -16.099  -7.743  10.541  59.162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.080e+02  1.847e+01  -5.846 9.27e-07 ***
## gdp_detrend_ts  1.939e-04  1.191e-05  16.274 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.67 on 38 degrees of freedom
## Multiple R-squared:  0.8745, Adjusted R-squared:  0.8712
## F-statistic: 264.8 on 1 and 38 DF,  p-value: < 2.2e-16
```

correcting serial correlation

```
acf(residuals(sm_fdf))
```

Series residuals(sm_fdf)



```
dwtest(sm_fdf)
```

```
##
## Durbin-Watson test
##
## data: sm_fdf
## DW = 0.69595, p-value = 4.022e-07
## alternative hypothesis: true autocorrelation is greater than 0
```

```
robust_se <- sqrt(diag(NeweyWest(sm_fdf, lag = 5, prewhite = FALSE)))
coeftest(sm_fdf, vcov = NeweyWest(sm_fdf, lag = 5, prewhite = FALSE))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.0798e+02 2.5222e+01 -4.2812 0.0001216 ***
## gdp_detrend_ts 1.9389e-04 1.7847e-05 10.8640 3.247e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Part 4 —

4.4.1 —

```
lnloanir<-log(data_houses$house_loan_interest)
data_houses$con_dwell<-data_houses$const_overall-data_houses$constr_nondwell

print(data_houses$con_dwell)
```

```
## [1] 4601 7466 6304 6372 4827 6905 6605 7347 6268 8179 8051 7698
## [13] 7316 8580 7819 7534 9279 9193 9695 9120 8830 9087 8243 7781
## [25] 8185 11785 12450 10896 10329 11456 8764 9675 9015 8013 8743 8016
## [37] 7762 8815 9390 8840
```

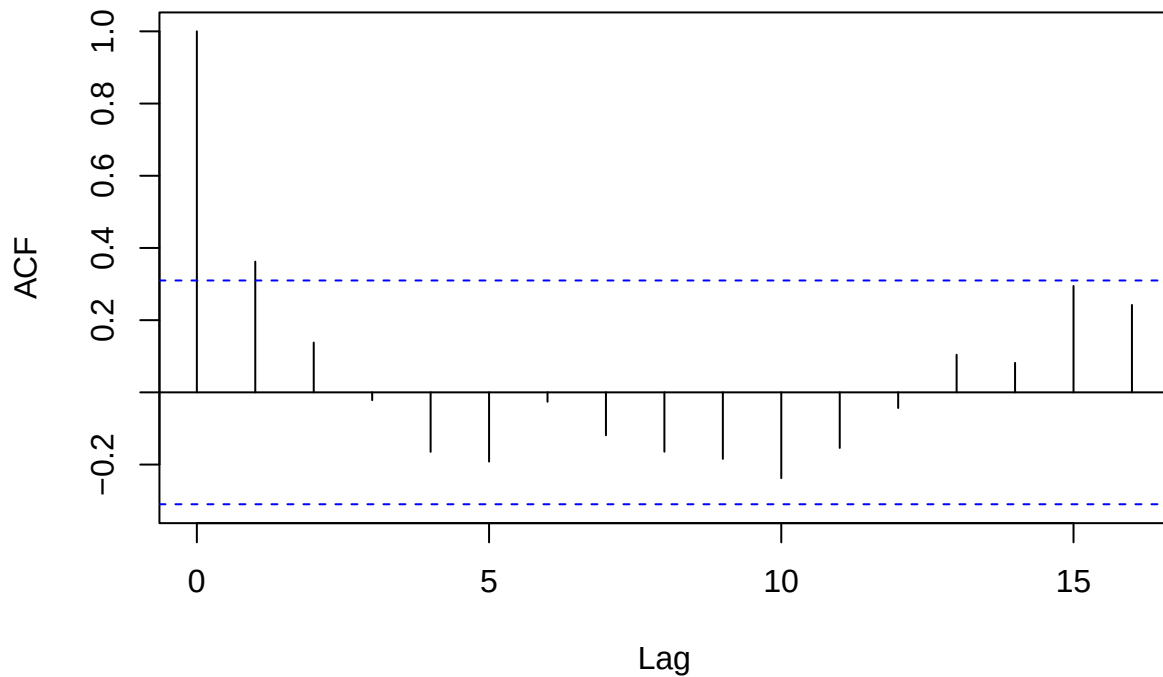
```
exp_in_model<-lm(data_houses$house_price_idx_2010~data_houses$nom_gdp+data_houses$employed_1000+data_ho
summary(exp_in_model)
```

```
##
## Call:
## lm(formula = data_houses$house_price_idx_2010 ~ data_houses$nom_gdp +
##     data_houses$employed_1000 + data_houses$con_dwell + lnloanir +
##     data_houses$house_loan_volume + data_houses$prod_price_idx_2021)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.169  -5.869  -2.128   6.604  22.763
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -4.381e+02  2.179e+02  -2.010  0.05263 .
## data_houses$nom_gdp    -5.362e-05  3.522e-05  -1.523  0.13738
## data_houses$employed_1000  2.952e-02  4.095e-02   0.721  0.47599
## data_houses$con_dwell    7.510e-03  1.313e-03   5.720 2.20e-06 ***
## lnloanir        -5.771e+01  2.065e+01  -2.795  0.00859 **
## data_houses$house_loan_volume  -6.524e-05  2.332e-04  -0.280  0.78138
## data_houses$prod_price_idx_2021  5.678e+00  8.548e-01   6.642 1.48e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.741 on 33 degrees of freedom
## Multiple R-squared:  0.978, Adjusted R-squared:  0.974
## F-statistic: 244.3 on 6 and 33 DF, p-value: < 2.2e-16
```

4.4.2 —

```
acf(residuals(exp_in_model))
```

Series residuals(exp_in_model)



```
dwtest(exp_in_model)
```

```
##  
## Durbin-Watson test  
##  
## data: exp_in_model  
## DW = 1.2609, p-value = 0.0007191  
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bptest(exp_in_model)
```

```
##  
## studentized Breusch-Pagan test  
##  
## data: exp_in_model  
## BP = 9.9794, df = 6, p-value = 0.1255
```

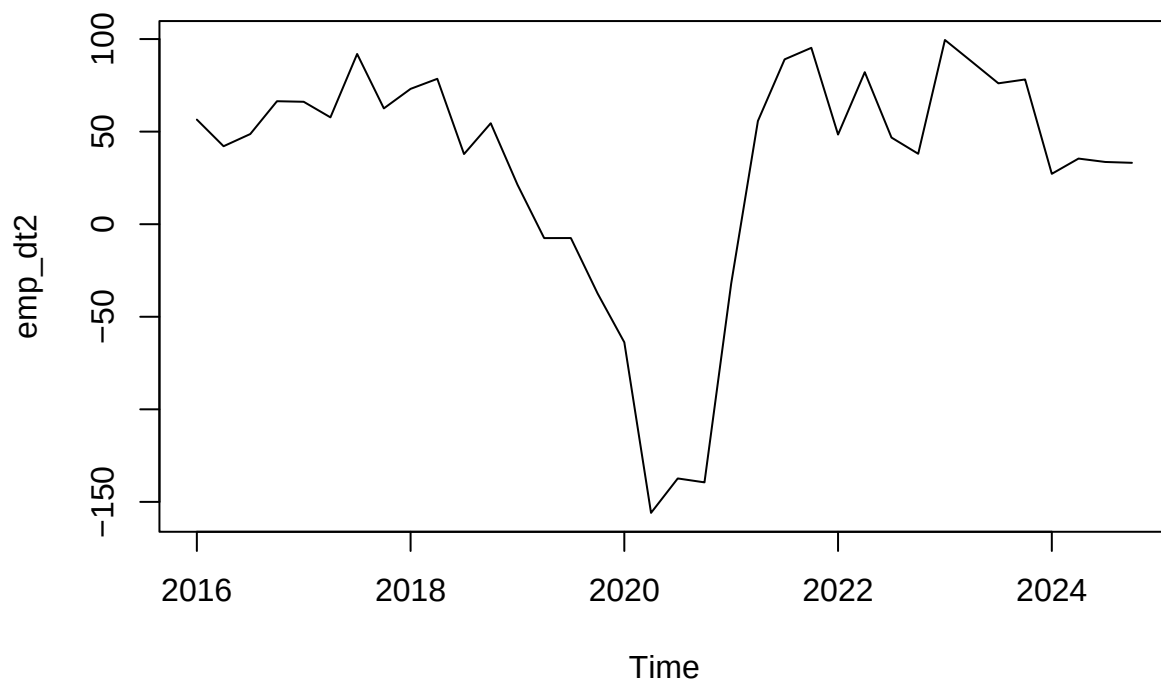
quarter correction

```
adf.test(data_houses$employed_1000)
```

```
##  
## Augmented Dickey-Fuller Test
```

```
##
## data: data_houses$employed_1000
## Dickey-Fuller = -1.2936, Lag order = 3, p-value = 0.8507
## alternative hypothesis: stationary
```

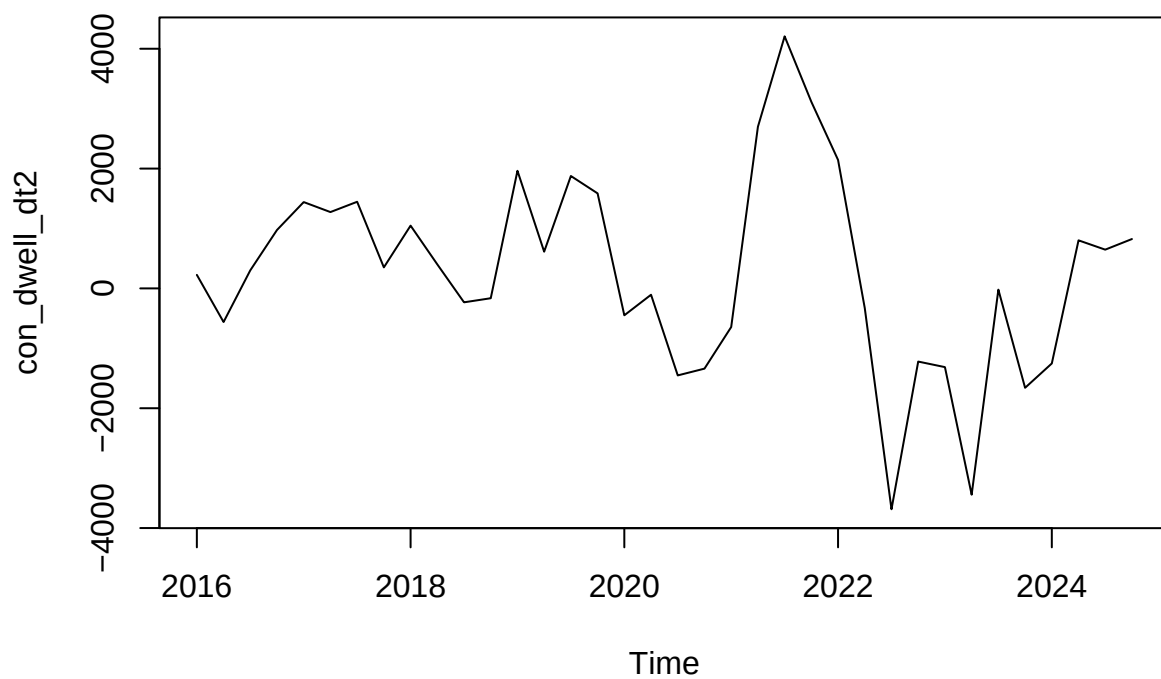
```
emp_ts <- ts(data_houses$employed_1000, start = c(2015, 1), frequency = 4)
emp_dt2 <- diff(emp_ts, lag = 4)
plot(emp_dt2)
```



```
adf.test(data_houses$con_dwell)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: data_houses$con_dwell
## Dickey-Fuller = -1.6522, Lag order = 3, p-value = 0.71
## alternative hypothesis: stationary
```

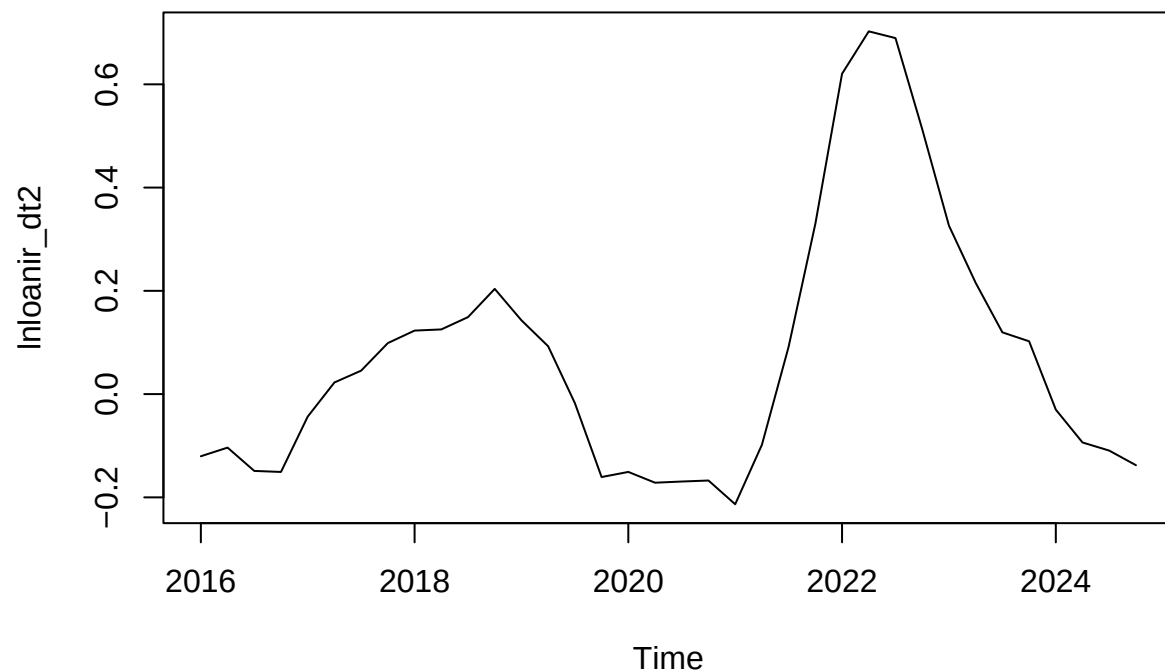
```
con_dwell_ts <- ts(data_houses$con_dwell, start = c(2015, 1), frequency = 4)
con_dwell_dt2 <- diff(con_dwell_ts, lag = 4)
plot(con_dwell_dt2)
```



```
adf.test(lnloanir)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: lnloanir  
## Dickey-Fuller = -2.6847, Lag order = 3, p-value = 0.3048  
## alternative hypothesis: stationary
```

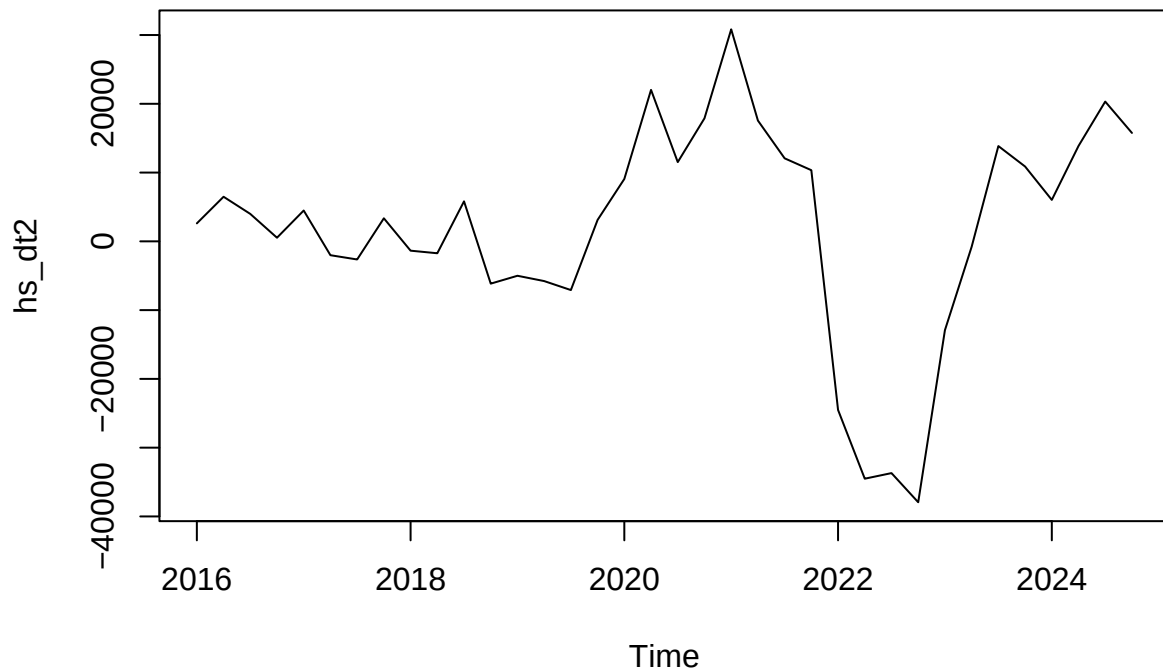
```
lnloanir_ts <- ts(lnloanir, start = c(2015, 1), frequency = 4)  
lnloanir_dt2 <- diff(lnloanir_ts, lag = 4)  
  
plot(lnloanir_dt2)
```



```
adf.test(data_houses$house_loan_volume)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: data_houses$house_loan_volume  
## Dickey-Fuller = -2.6197, Lag order = 3, p-value = 0.3302  
## alternative hypothesis: stationary
```

```
hlvl_ts <- ts(data_houses$house_loan_volume, start = c(2015, 1), frequency = 4)  
hs_dt2 <- diff(hlvl_ts, lag = 4)  
plot(hs_dt2)
```

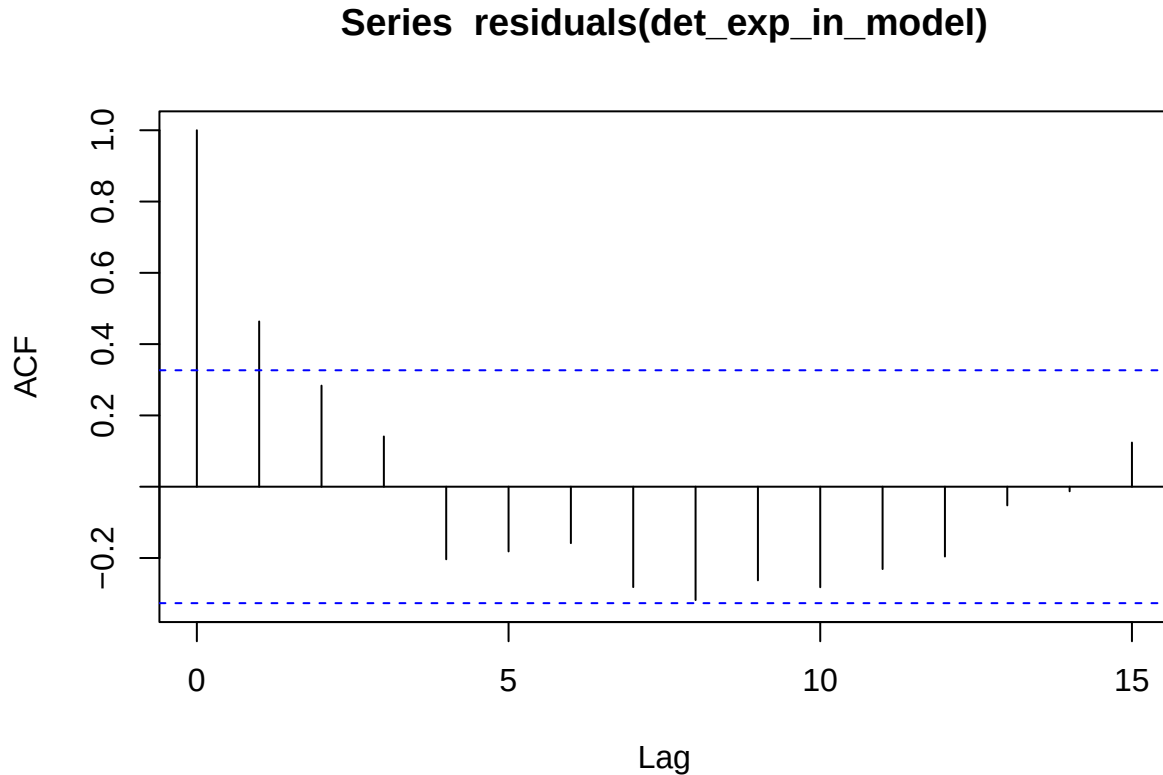


```
det_exp_in_model<-lm(hpi_detrend~gdp_detrend1+emp_dt2+con_dwell_dt2+lnloanir_dt2+hs_dt2)
summary(det_exp_in_model)
```

```
##
## Call:
## lm(formula = hpi_detrend ~ gdp_detrend1 + emp_dt2 + con_dwell_dt2 +
##     lnloanir_dt2 + hs_dt2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.990  -6.753   0.733   7.177  23.199
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.949e+01  4.881e+00   3.994 0.000389 ***
## gdp_detrend1 -4.956e-05  5.530e-05  -0.896 0.377218
## emp_dt2      -1.137e-01  4.514e-02  -2.519 0.017329 *
## con_dwell_dt2  7.090e-03  1.401e-03   5.062 1.96e-05 ***
## lnloanir_dt2  6.025e+01  1.788e+01   3.370 0.002080 **
## hs_dt2       2.355e-04  2.510e-04   0.938 0.355547
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.68 on 30 degrees of freedom
## Multiple R-squared:  0.5435, Adjusted R-squared:  0.4674
## F-statistic: 7.144 on 5 and 30 DF,  p-value: 0.0001661
```

robust standard errors

```
acf(residuals(det_exp_in_model))
```



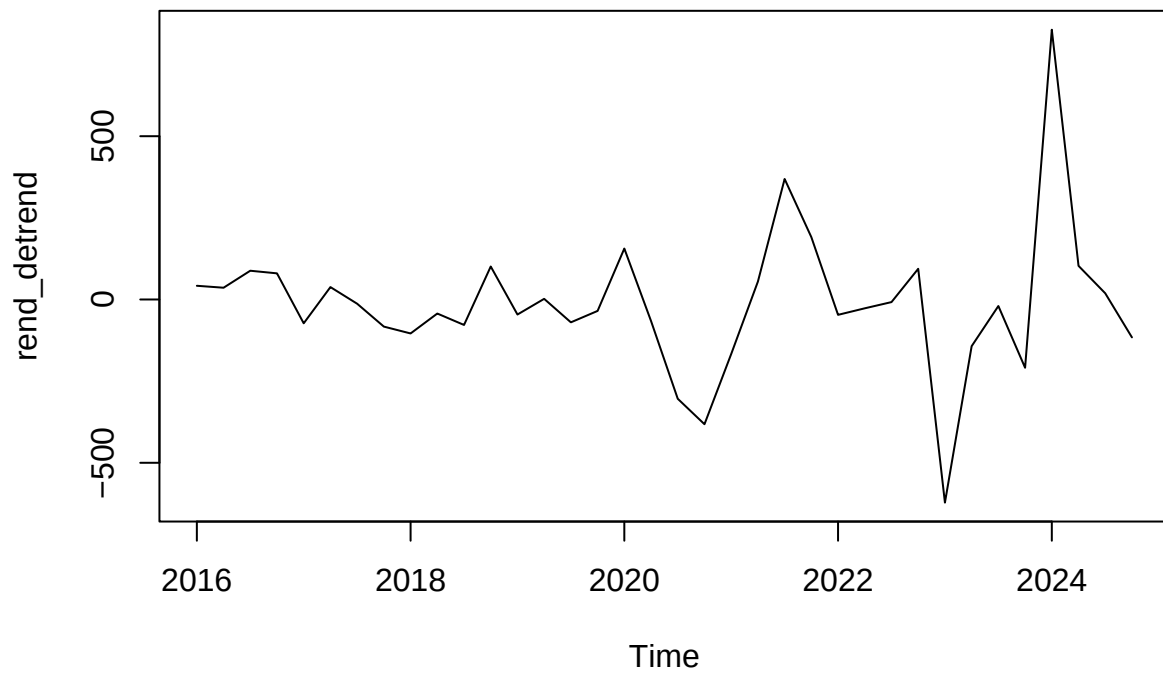
```
robust_se_1 <- sqrt(diag(NeweyWest(det_exp_in_model, lag = 2, prewhite = FALSE)))  
coeftest(det_exp_in_model, vcov = NeweyWest(det_exp_in_model, lag = 2, prewhite = FALSE))
```

```
##  
## t test of coefficients:  
##  
##           Estimate Std. Error t value Pr(>|t|)  
## (Intercept)  1.9492e+01 4.0722e+00  4.7866 4.254e-05 ***  
## gdp_detrend1 -4.9564e-05 4.8461e-05 -1.0228 0.3145907  
## emp_dt2      -1.1370e-01 3.7446e-02 -3.0363 0.0049181 **  
## con_dwell_dt2 7.0897e-03 1.6182e-03  4.3814 0.0001327 ***  
## lnloanir_dt2  6.0249e+01 2.0186e+01  2.9847 0.0056011 **  
## hs_dt2        2.3552e-04 3.4697e-04  0.6788 0.5024796  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

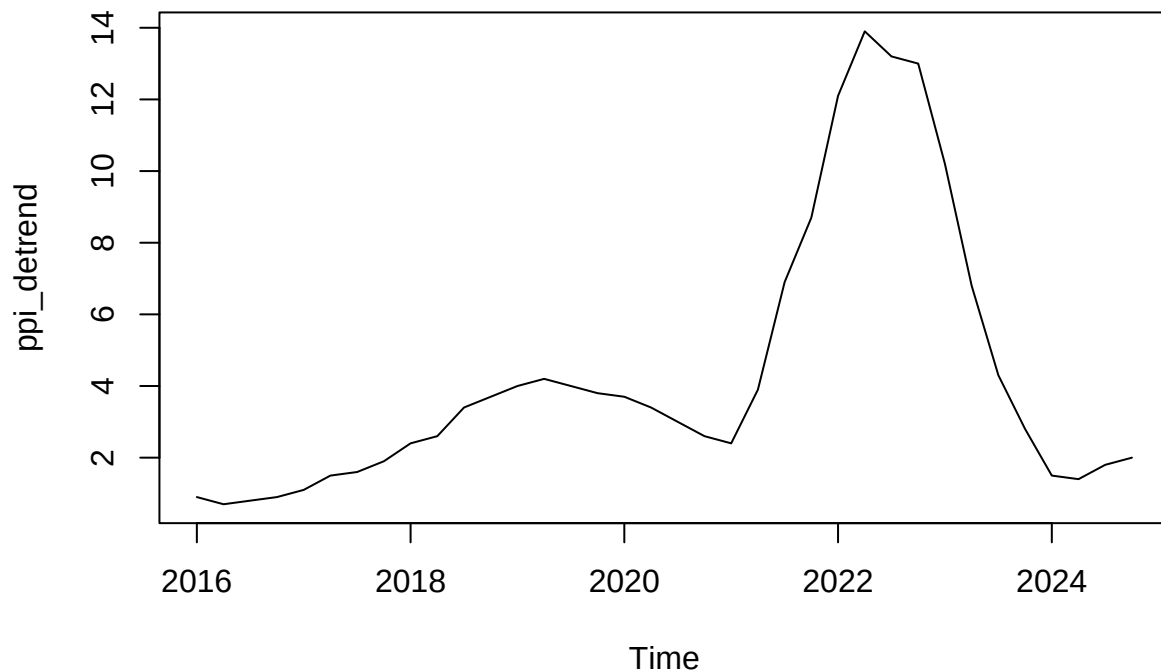
third model

```
data_houses$ren_dwell<-data_houses$renov_fam-data_houses$renov_apart
```

```
rd_ts <- ts(data_houses$ren_dwell, start = c(2015, 1), frequency = 4)
rend_detrend <- diff(rd_ts, lag = 4)
plot(rend_detrend)
```



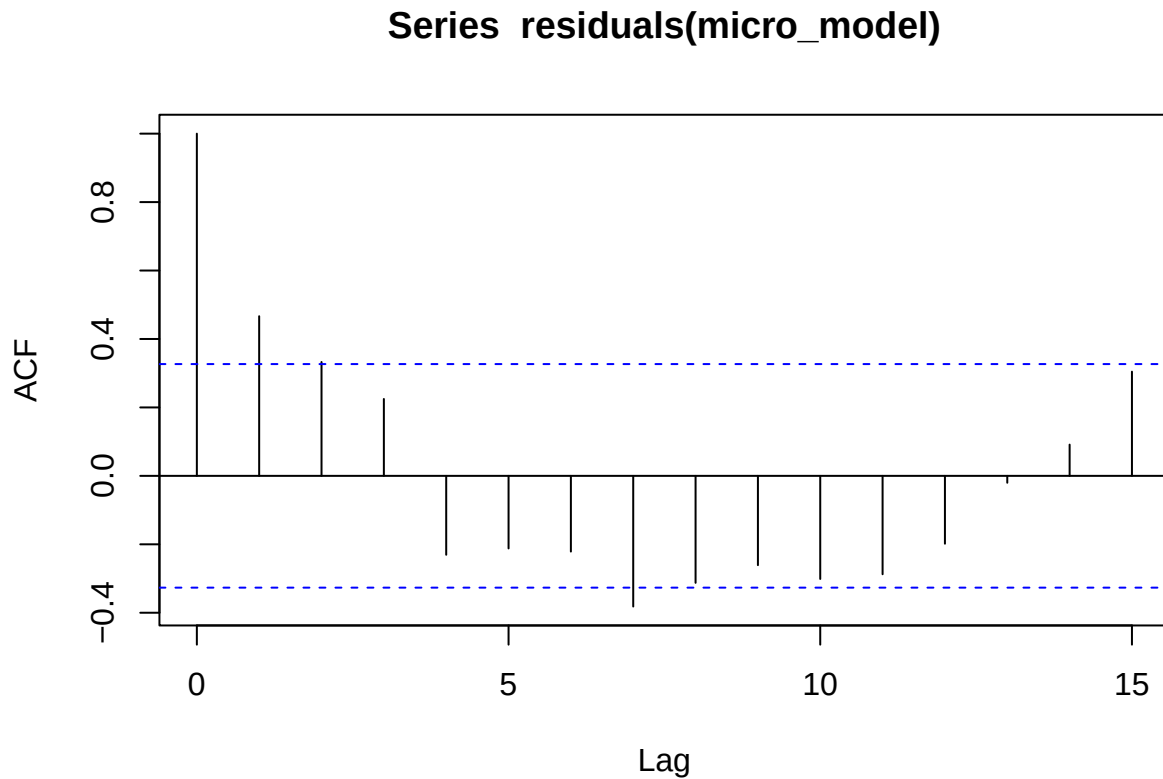
```
ppi_ts <- ts(data_houses$prod_price_idx_2021, start = c(2015, 1), frequency = 4)
ppi_detrend <- diff(ppi_ts, lag = 4)
plot(ppi_detrend)
```



```
micro_model<-lm(hpi_detrend~emp_dt2+con_dwell_dt2+lnloanir_dt2+rend_detrend+ppi_detrend)
summary(micro_model)
```

```
##
## Call:
## lm(formula = hpi_detrend ~ emp_dt2 + con_dwell_dt2 + lnloanir_dt2 +
##      rend_detrend + ppi_detrend)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -32.612  -6.954   1.313   6.695  22.179
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  14.598178   5.795579   2.519   0.0173 *
## emp_dt2      -0.125383   0.046563  -2.693   0.0115 *
## con_dwell_dt2  0.006730   0.001457   4.618 6.84e-05 ***
## lnloanir_dt2  29.798677  25.132296   1.186   0.2451
## rend_detrend   0.010919   0.010323   1.058   0.2986
## ppi_detrend    0.802772   1.457773   0.551   0.5859
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.76 on 30 degrees of freedom
## Multiple R-squared:  0.5376, Adjusted R-squared:  0.4606
## F-statistic: 6.977 on 5 and 30 DF, p-value: 0.0001984
```

```
acf(residuals(micro_model))
```



```
acf(residuals(micro_model))  
dwtest(micro_model)
```

```
##  
## Durbin-Watson test  
##  
## data: micro_model  
## DW = 1.0385, p-value = 0.0001129  
## alternative hypothesis: true autocorrelation is greater than 0
```

```
bptest(micro_model)
```

```
##  
## studentized Breusch-Pagan test  
##  
## data: micro_model  
## BP = 5.1197, df = 5, p-value = 0.4014
```

```
robust_se_mm <- sqrt(diag(NeweyWest(micro_model, lag = 3, prewhite = FALSE)))  
coeftest(micro_model, vcov = NeweyWest(micro_model, lag = 3, prewhite = FALSE))
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  14.5981776  5.1135571  2.8548 0.0077385 **
## emp_dt2      -0.1253828  0.0541085 -2.3172 0.0274993 *
## con_dwell_dt2 0.0067302  0.0016442  4.0932 0.0002954 ***
## lnloanir_dt2 29.7986773 27.9541105  1.0660 0.2949344
## rend_detrnd  0.0109185  0.0074489  1.4658 0.1531092
## ppi_detrnd   0.8027721  1.4751659  0.5442 0.5903321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```